instructions. When triggered either manually by command authorities or automatically through sensor alerts drones can be deployed in seconds to respond to fires, intrusions, industrial accidents, or natural disasters. Using real-time streaming over 5G/LTE, video and telemetry data are transmitted directly to emergency command centers, enabling faster situational awareness, decision-making, and coordinated response.

Nokia Drone Networks also integrates AI-based video analytics, allowing the system to detect anomalies such as unauthorized movement, structural damage, hazardous leaks, or evolving fire boundaries. This enables proactive and predictive emergency monitoring, particularly in ports, industrial factories, critical infrastructure zones, border environments, and urban security monitoring.

In addition to public safety, the system supports search and rescue, environmental inspections, and maritime patrols especially where ground access is limited or hazardous. The reliance on private cellular networks, rather than open public networks, enhances data security, communication resilience, and airspace safety, aligning the platform with European regulatory standards for unmanned aircraft operations.

Nokia Drone Networks demonstrates how network-enabled autonomy can transform drones into 24/7 aerial response units, expanding both the scale and reliability of drone-based surveillance and rescue missions. It also serves as a model for the integration of UAVs within smart-city, industrial safety, and national emergency management frameworks across Europe and beyond.

Figure 9.11 illustrates the Nokia Drone Networks platform, which enables fully automated drone operations using secure 5G/LTE communication links. The drone launches from an autonomous docking station that provides charging, data transfer, and mission updating without